

SmartBridge SKILL WALLET Scalability & Future Plan

Date	15 March 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current HDI prediction solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Flask single-threaded server handles limited concurrent prediction requests	Response time degrades significantly beyond 25 concurrent users	High
2	No persistent database; prediction history is not stored anywhere	Cannot review past predictions, build analytics, or track usage patterns	Medium
3	Model is trained on a fixed historical dataset; does not auto-retrain with new data	Predictions may drift in accuracy as global development indicators change over time	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Gunicorn worker on Render free tier; handles ~25 concurrent users	Deploy multiple Gunicorn workers; migrate to Docker + cloud auto-scaling (AWS/GCP) for higher concurrency
2	Data Storage	Flat CSV files; no database; no prediction logging	Integrate PostgreSQL to store prediction logs, user sessions and indicator inputs; enables audit trail and usage analytics

3	Performance	model.pkl loaded at startup; no caching for repeated identical inputs	Add Redis caching for duplicate input combinations; migrate to FastAPI for async endpoints to improve throughput
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S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	Security	No authentication; /predict endpoint is publicly accessible with no rate limiting	Add API key authentication for programmatic access; enforce HTTPS and add rate limiting via Nginx or a reverse proxy

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate a Gen-AI assistant (LLM) that explains the predicted HDI category in plain language and suggests policy interventions for low-scoring regions	1 month post-deployment	Improved actionability of results; non-technical users can understand and act on predictions
Phase 3	Connect to live World Bank / UNDP open data APIs so indicator fields are auto-populated with the latest available country data	1 month post-deployment	Eliminates manual data entry; predictions always use up-to-date indicators
Phase 4	Build a comparative analytics dashboard showing HDI category trends over time across countries and regions, with filtering by indicator	1 month post-deployment	Enables longitudinal development analysis for policymakers and researchers; extends the tool beyond single-prediction use